

PHASE 2 · SHORT VERSION

# Reproducing EfficientNet-B0 for malaria detection

From the paper's own code to all 10 folds: results, findings and what we test next

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# What we did after Phase 1, and what we found



## Phase 1 • reviewed 11 studies

Accuracy on the NIH cells is reported at 96–99.9 %, but most studies split images at random, so one patient's cells sit in both training and testing.



## Phase 2 • reproduced the base paper

Marques et al. (2022): EfficientNet-B0 + a 10-fold ensemble. We rebuilt it from the authors' own code and trained all 10 folds.



## Next • test it fairly

H1: does accuracy drop once slide-level leakage is removed? H2: does YUV stain normalisation win it back?

**97.55 %**

**mean fold accuracy**

paper 97.56 % ·  $\Delta$  -0.01 pp

**97.62 %**

**10-model ensemble**

paper 98.29 % ·  $\Delta$  -0.67 pp

**99.73 %**

**ensemble ROC-AUC**

paper 99.76 % ·  $\Delta$  -0.03 pp

**≈ 11 min**

**per fold on an RTX 4090**

paper ≈ 4 h 46 min per fold

All 10 folds within 1 percentage point of the paper. Every metric is computed from the saved predictions in `results/A0_10fold_paper/`.

# Why EfficientNet-B0 with a 10-fold ensemble



## Open, executable code

The authors published their executed Jupyter notebook as a supplementary file, so a faithful rebuild is possible.



## Strict validation

Stratified 10-fold cross-validation plus a separate hold-out set that no model trains on.



## Light backbone

4.2 M parameters in total: the smallest EfficientNet, close to our deployability goal.



## Strong, not at the ceiling

98.29 % leaves room to see whether a change helps or hurts.

### THE PAPER'S TARGETS (from its own printed results)

**97.56 %**

mean fold accuracy (Table 5)

**97.69 %**

single model on hold-out (Table 6)

**98.29 %**

ensemble accuracy

**99.76 %**

ensemble ROC-AUC

# The authors' notebook differs from the paper's text

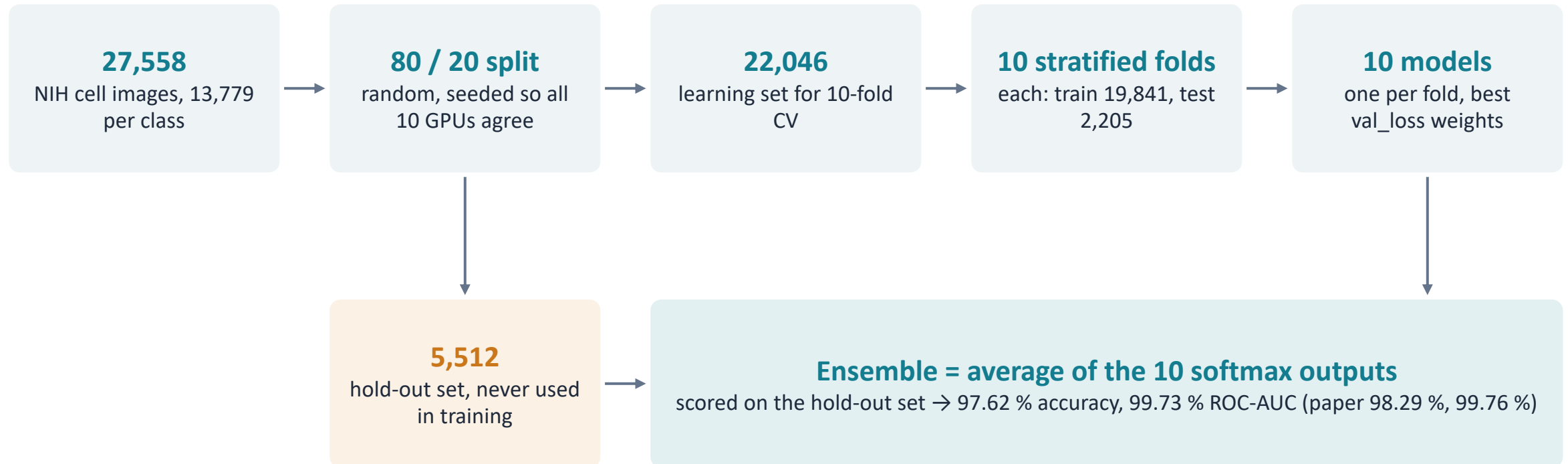
| Setting            | Paper's code (used now)                                           | Our first reproduction (from the text) |
|--------------------|-------------------------------------------------------------------|----------------------------------------|
| Pretrained weights | noisy-student · efficientnet 1.1.1 (qubvel)                       | ImageNet · tf.keras.applications       |
| Classifier head    | GAP → 128 → 64 → 32 → 2 softmax, dropout 0.3                      | GAP → dropout 0.2 → 1 sigmoid          |
| Loss               | categorical cross-entropy, label smoothing 0.1                    | binary cross-entropy                   |
| Schedule           | 33 epochs, LR × 0.5 on plateau, keep best val_loss                | ≤ 15 epochs, LR × 0.1, early stopping  |
| Batch size         | 16 (1,240 steps per epoch)                                        | 128 (64 per GPU, 2 GPUs)               |
| Augmentation       | full albumentations pipeline (noise, blur, distortions, CLAHE...) | flips + 90° rotations                  |

Source: Springer supplementary file MOESM1, the authors' executed notebook (30 Jan 2021). Our notebook now follows it line by line; 32 automated tests check the recipe.



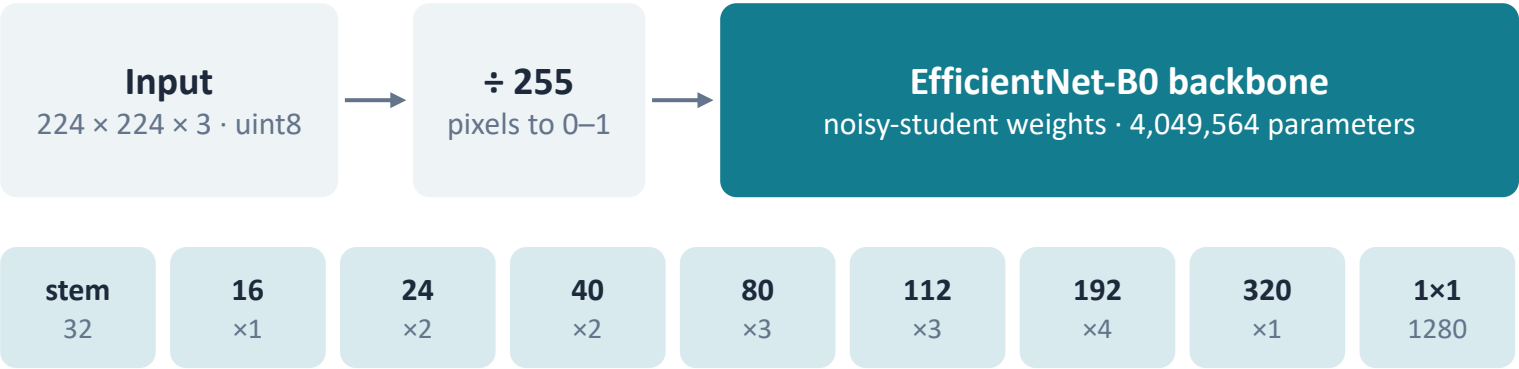
Two quirks in the paper itself: its code evaluates on augmented images, and its text swaps precision and recall (true recall 97.74 %, precision 98.82 %).

# One split, ten fold models, one ensemble



Every fold model is also scored on its own CV test part (the paper's Table 5) and on the hold-out set (Table 6). Each fold stores a fingerprint of the split (a0cc112e...): all 10 folds match.

# EfficientNet-B0 with the paper's dense head



Inside the backbone: a stem convolution, 16 MBConv blocks in 7 stages (output channels, repeats), and a  $1 \times 1$  convolution  $\rightarrow 7 \times 7 \times 1280$  features.



Output: P(parasitized), P(uninfected). Loss: categorical cross-entropy with label smoothing 0.1.

| Part                      | Parameters       |
|---------------------------|------------------|
| EfficientNet-B0 backbone  | 4,049,564        |
| Dense head                | 174,370          |
| <b>Total</b>              | <b>4,223,934</b> |
| Trainable                 | 4,181,918        |
| Non-trainable (BatchNorm) | 42,016           |

Identical to the paper's Table 2, digit for digit, checked by an automated test.

Noisy-student weights: EfficientNet-B0 pre-trained on ImageNet plus self-training on unlabelled images (Xie et al., 2020). The paper loads them from the efficientnet package; so do we.

# How to read the metrics

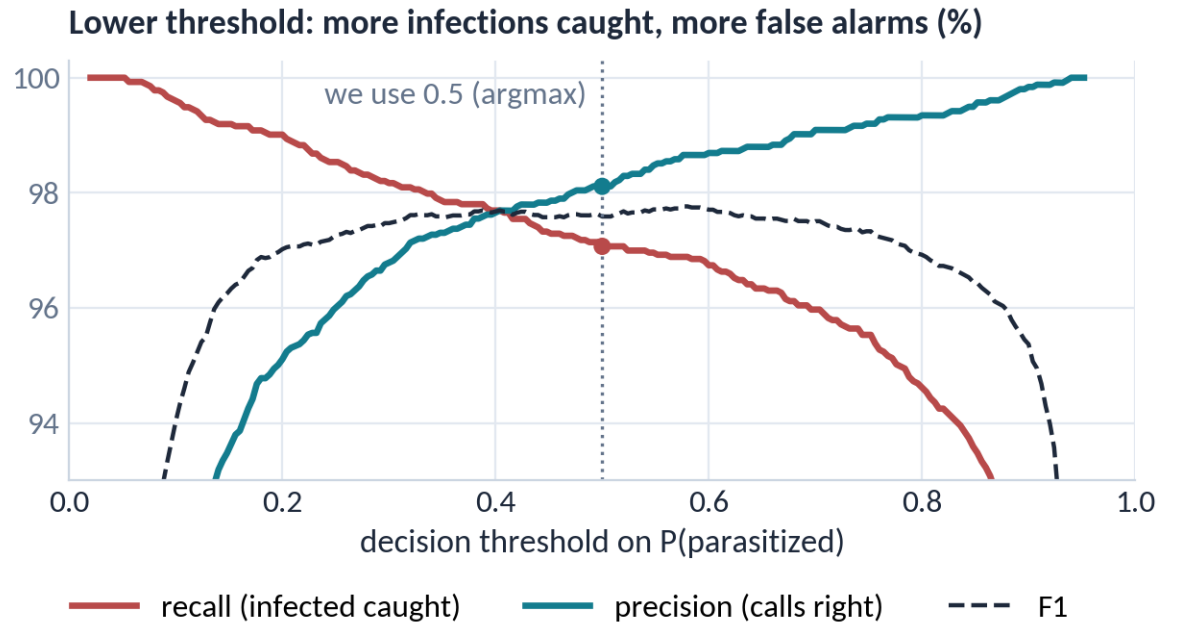
|                                  | model says<br>PARASITIZED              | model says<br>UNINFECTED              |
|----------------------------------|----------------------------------------|---------------------------------------|
| truth:<br>parasitized<br>(2,730) | <b>TP = 2,650</b><br>infected, caught  | <b>FN = 80</b><br>infected, missed    |
| truth:<br>uninfected<br>(2,782)  | <b>FP = 51</b><br>healthy, false alarm | <b>TN = 2,731</b><br>healthy, cleared |

Accuracy =  $(TP + TN) / \text{all} = 5,381 / 5,512 = 97.62 \%$

**Recall (sensitivity)** =  $TP / (TP + FN)$ : infected cells caught · 97.07 %

**Precision** =  $TP / (TP + FP)$ : 'parasitized' calls that are right · 98.11 %

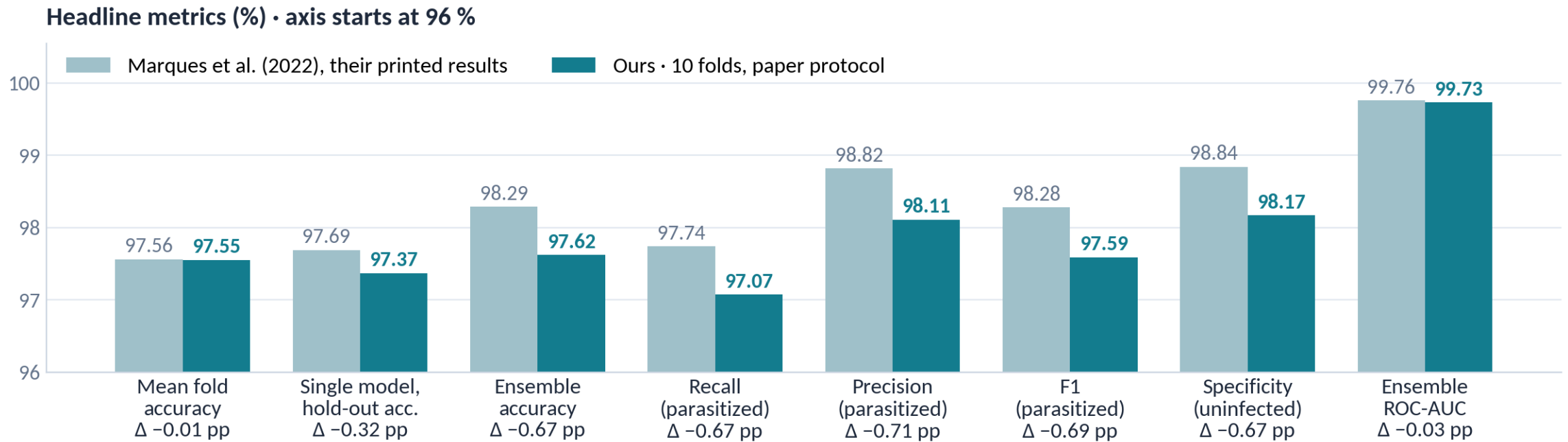
**F1** = harmonic mean of precision and recall · 97.59 %



**ROC-AUC**: chance that an infected cell outscores a healthy one · 99.73 %

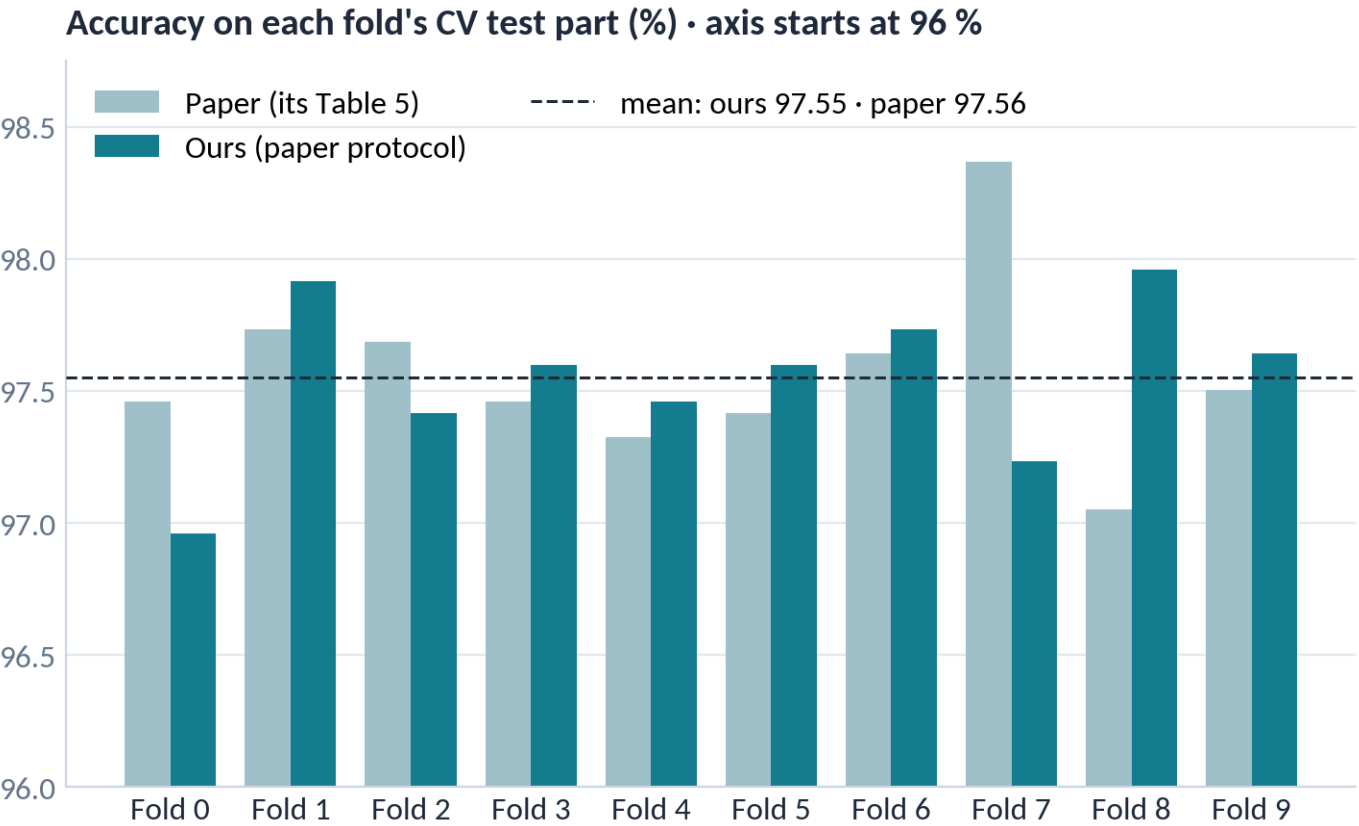
**pp (percentage point)** = plain difference of two percentages:  $97.62 - 98.29 = -0.67 \text{ pp}$

# Paper vs. ours on the 5,512-image hold-out set



All within our 1 pp success criterion; mean fold accuracy matches to 0.01 pp. On clean (un-augmented) images the ensemble scores 97.64 %, so the paper's augmented evaluation changes little.

# Ten consistent folds, and where the ensemble errs



## Ensemble on 5,512 hold-out cells

|             | Pred. parasitized | Pred. uninfected |
|-------------|-------------------|------------------|
| Parasitized | 2,650             | 80               |
| Uninfected  | 51                | 2,731            |

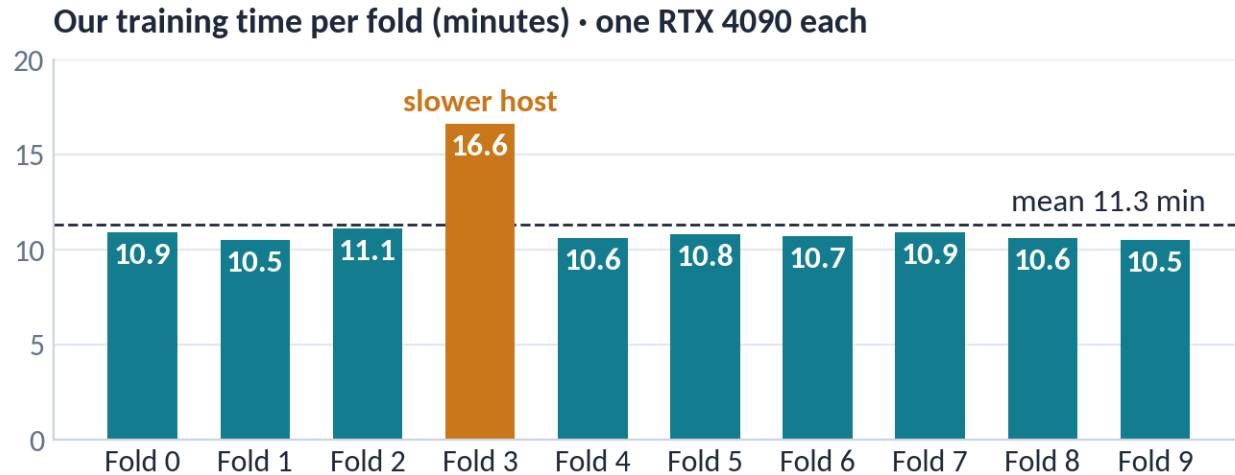
Paper: 62 missed, 32 flagged. Recall (catching infected cells) is our priority metric.

# Stable training, near-perfect ranking



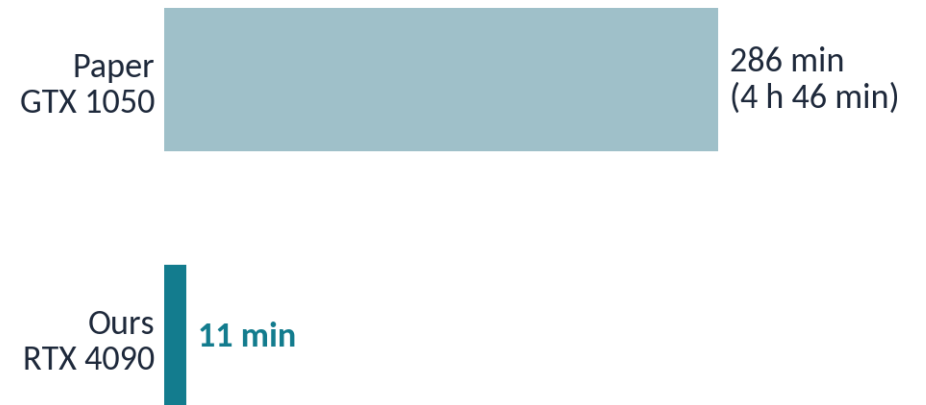
Validation accuracy settles near 97.5 % after about 10 epochs; the learning rate halves on plateaus and the lowest-val\_loss weights are kept. The ensemble's ROC hugs the top-left corner: AUC 99.73 % (paper 99.76 %).

# Ten GPUs in parallel: about 11 minutes per fold



10 RunPod pods (Secure Cloud, RTX 4090, 16 vCPU each), one fold per pod, all training at the same time. Fold 3 landed on a slower host (16.6 min).

Training time per fold:  $\approx 25\times$  faster



$\approx 25\times$   
faster per fold

$\approx \$4.8$   
total cloud cost

**1 d 23 h**  
the paper's full run



The 10 trained models are published as a GitHub release, so nobody has to retrain to use or evaluate them.

# Two proposed changes, one variable at a time



## H1 · VALIDATION

### Leakage-free, slide-grouped split

#### CHANGE

Split by slide/patient (StratifiedGroupKFold) instead of random images. Model unchanged.

#### WE EXPECT

Accuracy and MCC fall below the image-split baseline, and fold-to-fold spread widens.



## H2 · PREPROCESSING

### YUV stain normalisation (Umer [4])

#### CHANGE

RGB → YUV → histogram-equalise Y → RGB before EfficientNet-B0. Grouped split kept fixed.

#### WE EXPECT

Higher recall and MCC than the H1 model on unseen slides; little change on the random split.

| ID | Split         | Preprocessing | Answers                              | Status            |
|----|---------------|---------------|--------------------------------------|-------------------|
| A0 | image (paper) | none          | baseline: does it match the paper?   | done ✓ (10 folds) |
| A1 | slide-grouped | none          | H1: leakage gap (A0 – A1)            | next              |
| A2 | slide-grouped | YUV + hist-eq | H2: gain on unseen slides (A2 – A1)  | next              |
| A3 | image (paper) | YUV + hist-eq | control: is the gain slide-specific? | next              |

## CONCLUSION

# Where we stand



### Reproduced

The paper's recipe, rebuilt from its own code, lands within 1 pp on every metric (mean fold 97.55 vs 97.56 %).



### Corrected

The paper evaluates on augmented images and swaps precision and recall in its text; we report both views.



### Fast and shareable

10 folds in parallel,  $\approx 11$  min each instead of  $\approx 4$  h 46 min; the trained models are published.



### Next

A1–A3 test whether 97–98 % holds on unseen slides (H1) and whether stain normalisation helps (H2).

**Thank you · questions welcome**

[github.com/amathziah/malaria-detection](https://github.com/amathziah/malaria-detection)